Business Summary Report
Predictive Delinquency Insights

1.Predictive Insights:

Based on exploratory data analysis (Task 1) and the predictive modeling framework (Task 2), the following key insights were identified:

Top 3 Risk Factors

- Payment History – Customers with 2+ missed payments in the past 6 months are over 3x more likely to become delinquent.
- Credit Utilization – Customers consistently using >70% of available credit show significantly higher risk.
- Debt-to-Income Ratio (DTI) – A DTI >0.5 strongly correlates with repayment difficulties.

High-Risk Customer Segments

- Younger customers (<30) with limited credit history and high utilization.
- Customers with unstable income or frequent job changes.
- Customers showing sudden increases in new credit applications.

These predictors allow Geldium to identify at-risk groups early and design proactive outreach strategies.

2. Recommendation Framework

SMART Recommendation

"Pilot a 6-week proactive SMS/email campaign targeting high-risk customers with >70% credit utilization and 2+ missed payments, aiming to reduce 30+ day delinquency by 12%."

Rationale

- Specific: Focuses on clearly defined at-risk groups.
- Measurable: Tracks delinquency reduction percentage.
- Actionable: Uses existing customer communication channels.
- Relevant: Directly addresses top predictors of delinquency.
- Time-bound: 6-week pilot ensures timely evaluation.

This approach balances business efficiency with customer support, enabling Geldium to reduce losses while maintaining customer trust.

3. Ethical and Responsible AI Considerations

Fairness Risks & Mitigation

- Bias in demographic segments – Risk that younger or lower-income customers may be unfairly flagged. *Mitigation:* Regular fairness audits and removal of proxy variables (e.g., ZIP code).
- Over-reliance on historical patterns – Past biased lending practices could be reinforced. *Mitigation:* Monitor model outputs across demographics using fairness metrics (demographic parity, equalized odds).

Explainability

- Use simple models (e.g., decision trees/logistic regression) or SHAP values for complex models to explain "why" a customer is flagged.
- Translate outputs into plain language summaries for non-technical stakeholders (e.g., "This customer is high risk due to consistently missing payments and using more than 70% of available credit").

Responsible Decision-Making

- Recommendations ensure actions are proportionate to risk (e.g., reminders before harsh measures).
- Customer trust is preserved through transparency and fairness.
- Model use is aligned with Geldium's strategy of reducing delinquency responsibly.

✅ This report provides the Head of Collections with a clear view of high-risk factors, actionable next steps, and ethical safeguards to ensure AI-driven decision-making is both effective and fair.